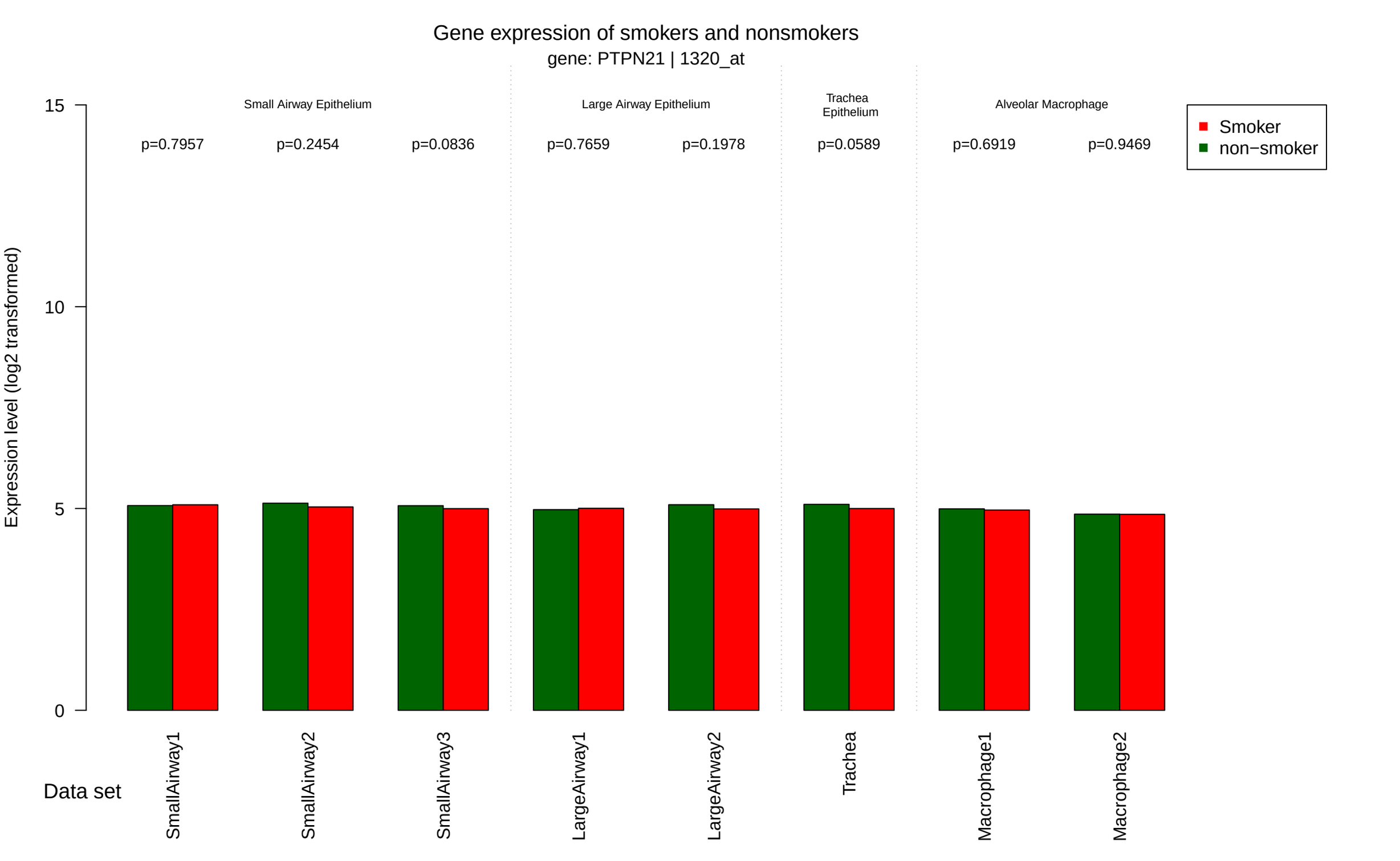
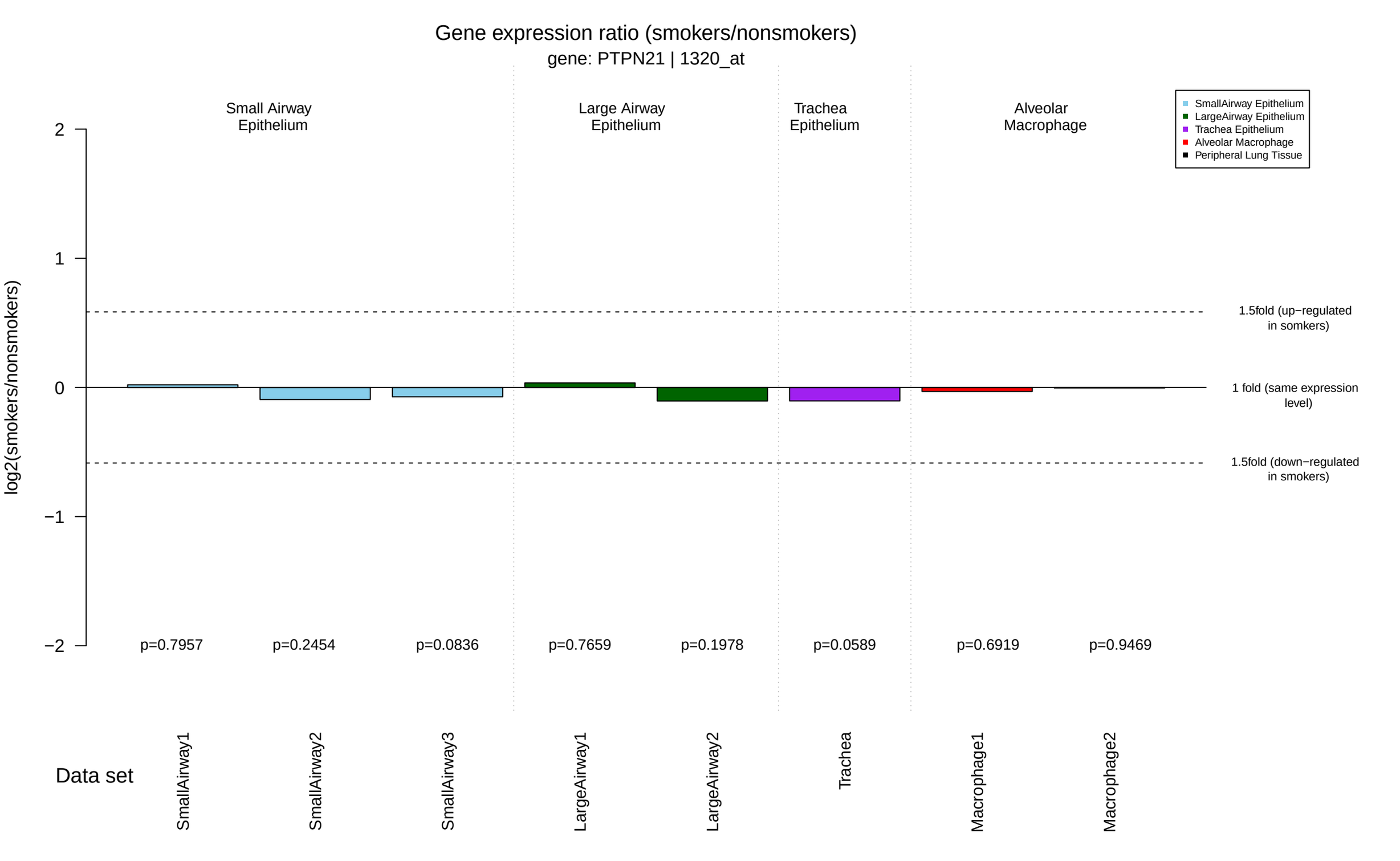
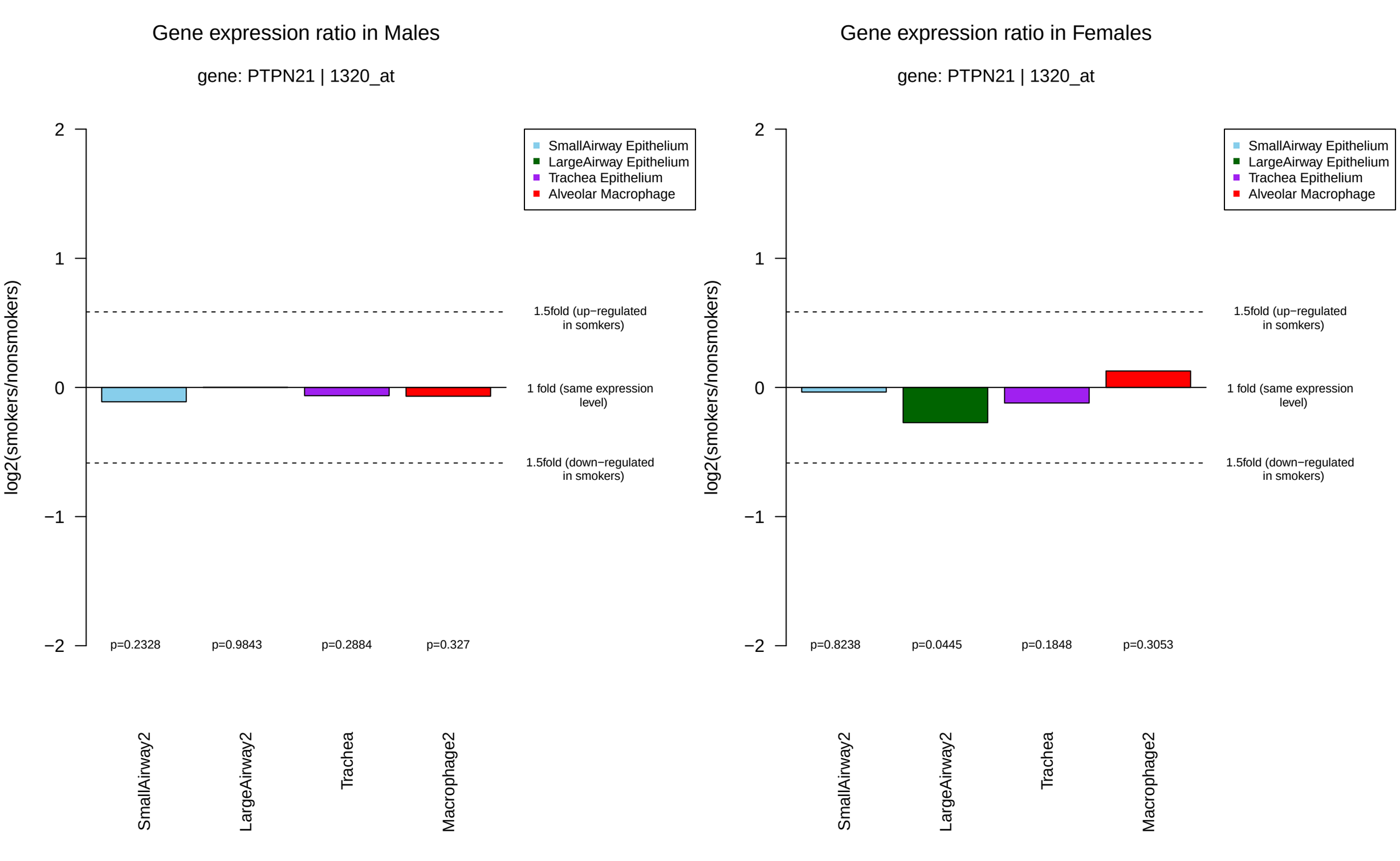
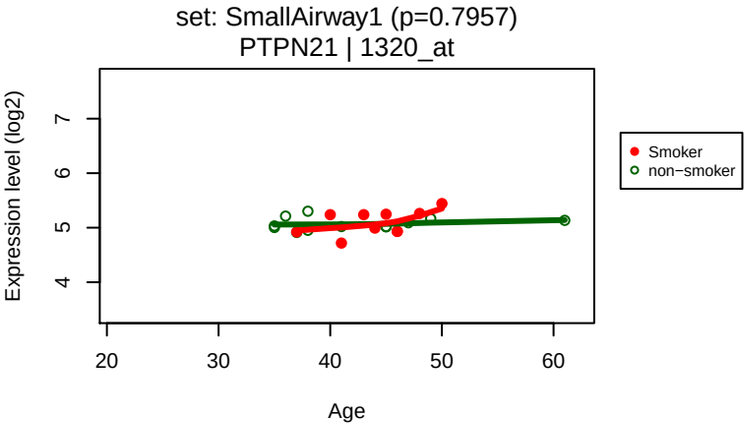
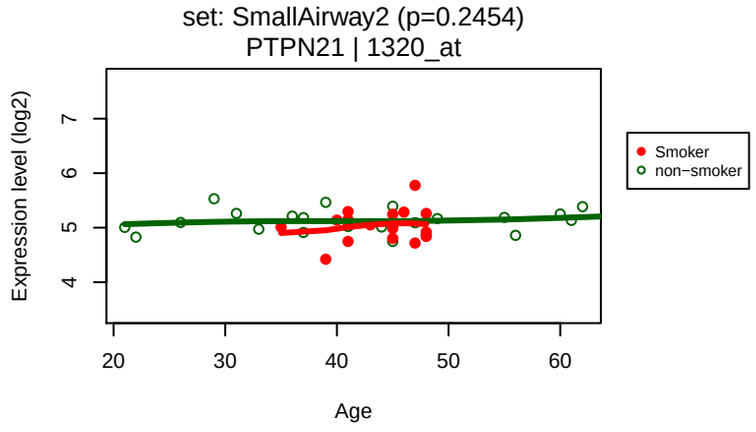
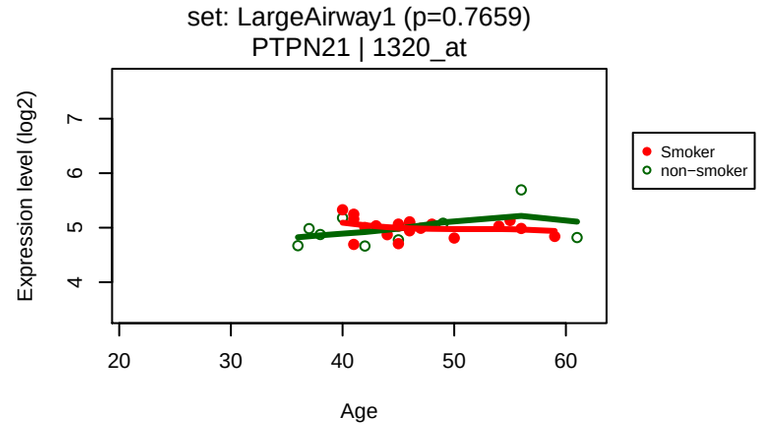

Gene expression of smokers and nonsmokers
gene: PTPN21 | 1320_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7957
p=0.2454
p=0.0836
p=0.7659
p=0.1978
p=0.0589
p=0.6919
p=0.9469
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

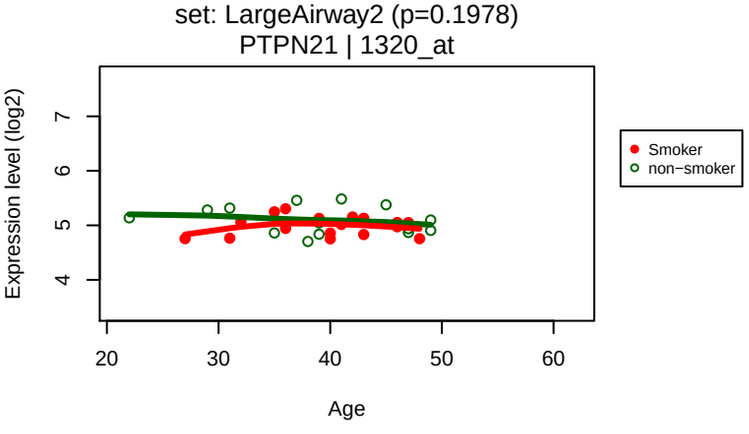

Gene expression ratio (smokers/nonsmokers)
gene: PTPN21 | 1320_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7957
p=0.2454
p=0.0836
p=0.7659
p=0.1978
p=0.0589
p=0.6919
p=0.9469
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

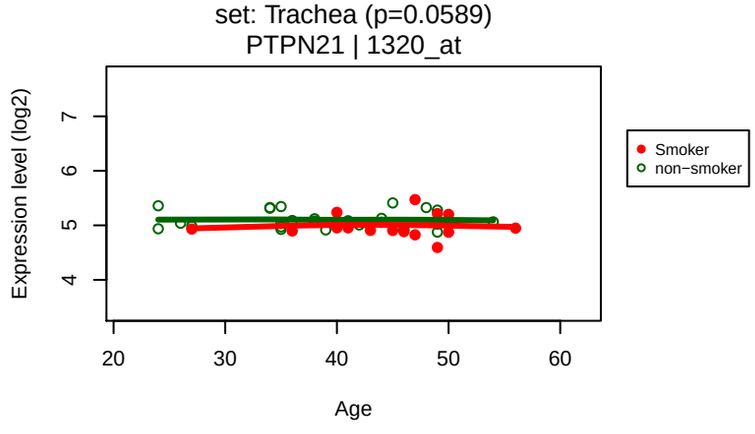

Gene expression ratio in Males
gene: PTPN21 | 1320_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2328
p=0.9843
p=0.2884
p=0.327
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTPN21 | 1320_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8238
p=0.0445
p=0.1848
p=0.3053
SmallAirway2
LargeAirway2
Trachea
Macrophage2

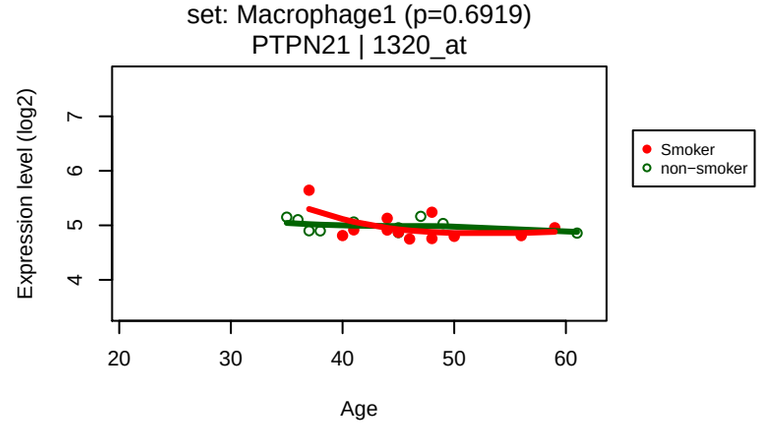

set: SmallAirway1 (p=0.7957)
PTPN21 | 1320_at
Expression level (log2)
Age
Smoker
non−smoker

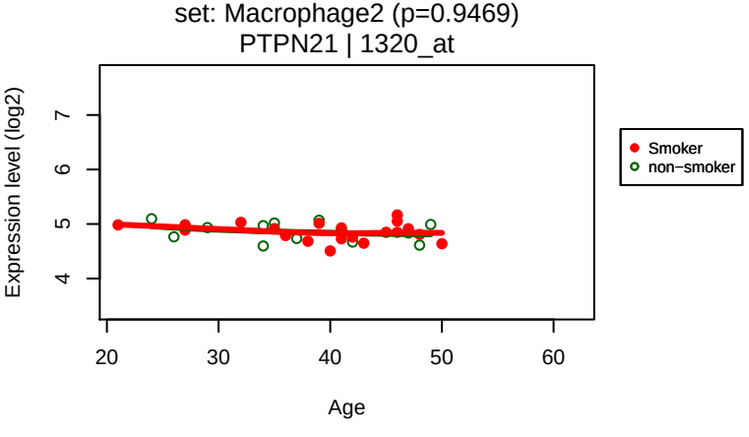

set: SmallAirway2 (p=0.2454)
PTPN21 | 1320_at
Expression level (log2)
Age
Smoker
non−smoker

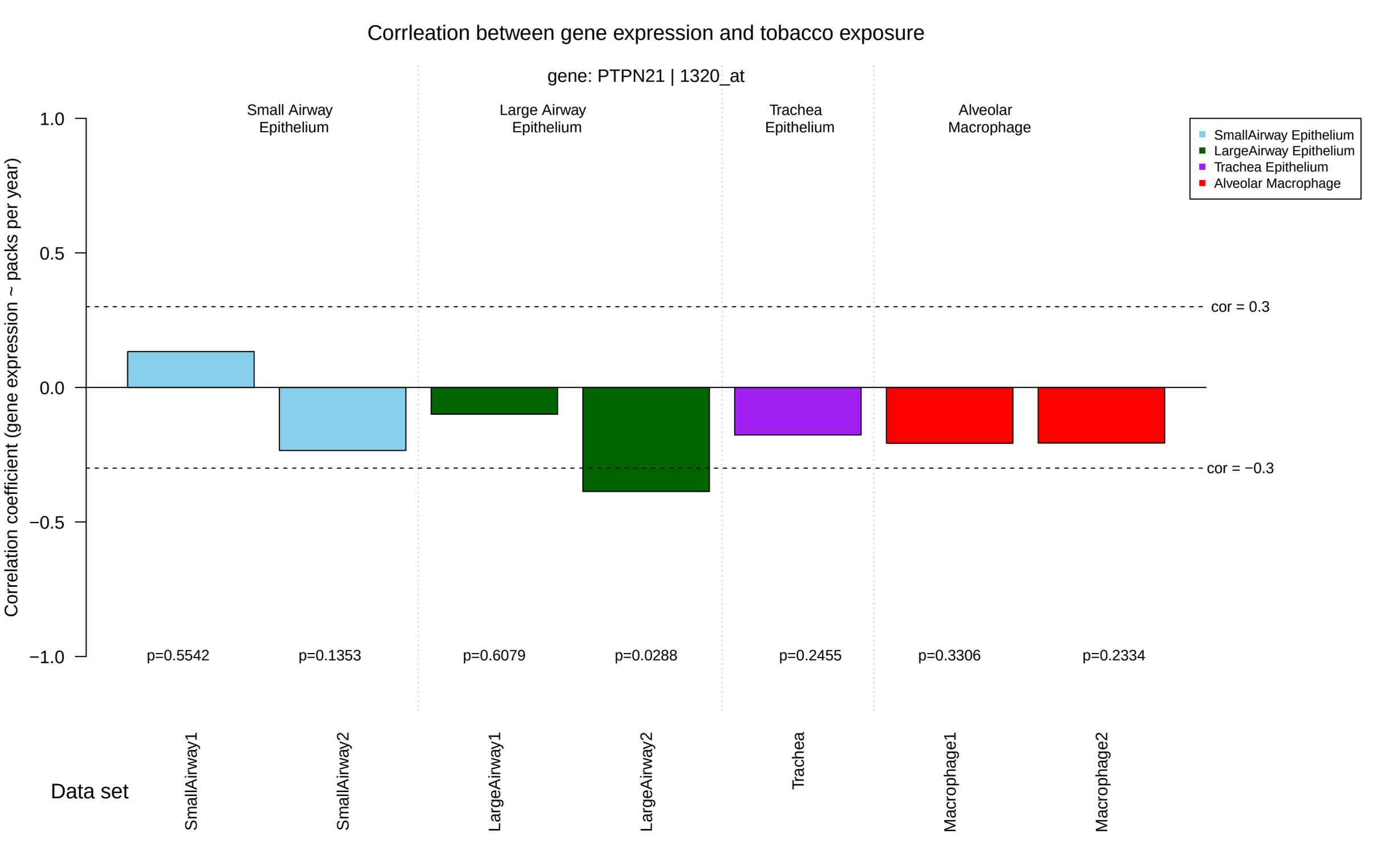

set: LargeAirway1 (p=0.7659)
PTPN21 | 1320_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1978)
PTPN21 | 1320_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0589)
PTPN21 | 1320_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6919)
PTPN21 | 1320_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9469)
PTPN21 | 1320_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTPN21 | 1320_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5542
p=0.1353
p=0.6079
p=0.0288
p=0.2455
p=0.3306
p=0.2334
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2